Identifying the telescoping continued-fraction coefficients

MathDB 354157: the stabilized coefficients equal the Stirling coefficients

Principia Math In Progress

12 September 2026

Task 1. Prove coefficient matching

Bhatnagar and Rajkumar's Conjecture 6.2 in [1], p. 9, asks whether the coefficients obtained by their telescoping algorithm agree with those of the classical continued fraction for the Stirling remainder. The algorithm first fixes a Legendre approximation order k, so denote its output by $a_j^{(k)}$. The sequence in the conjecture is the stabilized sequence described on p. 7 and displayed in (6.1). We prove its stabilization and identification together:

Theorem 1. *Let b_j be the classical Stirling continued-fraction coefficients. For every $m \geq 1$ and $k \geq m+1$, the source's successive leading-degree cancellations are possible through step m and give*

$$a_j^{(k)} = b_j \qquad (1 \leq j \leq m).$$

Consequently the stabilized sequence a_j satisfies $a_j = b_j$ for every j, resolving Conjecture 6.2.

The finite-k convention matters: for example, $a_2^{(2)} = 1/18$, whereas the stabilized value is $b_2 = 1/30$. The theorem concerns exactly the stabilization region $k \geq j+1$ specified in the source.

The two formal expansions

Let

$$r(x) = \log\Gamma(x+1) - (x+\tfrac{1}{2})\log x + x - \tfrac{1}{2}\log(2\pi), \qquad x > 0.$$

Its formal Stirling expansion and exact difference are

$$\mathcal{R}(x) = \sum_{\ell \geq 1} \frac{B_{2\ell}}{2\ell(2\ell-1)x^{2\ell-1}}, \tag{1}$$

$$r(x) - r(x+1) = \varepsilon(x) := (x+\tfrac{1}{2})\log(1+1/x) - 1. \tag{2}$$

Thus the formal expansion of ε is $\mathcal{R}(x) - \mathcal{R}(x+1)$. All formal O-statements below refer to Laurent expansions at infinity.

The classical coefficients $b_j > 0$ appearing in the conjecture are characterized by the formal continued fraction

$$\mathcal{R}(x) = \cfrac{b_1}{x + \cfrac{b_2}{x + \cfrac{b_3}{x + \ddots}}}.$$

In particular, if $B_m(x) = \mathrm{K}_{j=1}^{m} \frac{b_j}{x}$, then

$$\mathcal{R}(x) - B_m(x) = O(x^{-2m-1}). \tag{3}$$

These are the defining formal coefficients of the classical Stieltjes expansion recalled in [1, Section 6]. Only this formal property and nonvanishing of b_j are needed.

For completeness, the positive Stieltjes measure underlying these facts follows by integrating Binet's formula by parts:

$$\frac{r(\sqrt{z})}{\sqrt{z}} = \int_0^\infty \frac{\rho(t)}{z+t}\,dt, \qquad \rho(t) = \frac{-\log(1-e^{-2\pi\sqrt{t}})}{2\pi\sqrt{t}} > 0.$$

Its moments are $\mu_n = |B_{2n+2}|/[(2n+2)(2n+1)]$. Every ordinary and shifted Hankel moment determinant is positive, because its matrix is the Gram matrix of monomials for respectively $\rho(t)\,dt$ and $t\rho(t)\,dt$, both with infinite support. The classical Stieltjes correspondence therefore gives strictly positive, nonterminating continued-fraction coefficients. No convergence assertion is used to infer (3).

Accuracy of the finite Legendre input

Let P_k be the Legendre polynomial and P_k^* its associated numerator polynomial, with the normalization of [1, Section 2]. Put

$$E_k(x) = \frac{(2x+1)P_k^*(2x+1)}{P_k(2x+1)} - 1 = \frac{f_k^*(z)}{f_k(z)}, \qquad z = x(x+1).$$

For $k \geq 2$, $\deg f_k = d_k$ and $\deg f_k^* = d_k - 1$, after the usual cancellation when k is odd.

Lemma 1. *For each $k \geq 1$,*

$$\varepsilon(x) - E_k(x) = O(x^{-2k}).$$

Proof. The associated numerator has the integral expression

$$P_k^*(t) = \frac{1}{2}\int_{-1}^{1} \frac{P_k(t) - P_k(s)}{t-s}\,ds.$$

Consequently

$$\frac{1}{2}\log\frac{t+1}{t-1} - \frac{P_k^*(t)}{P_k(t)} = \frac{1}{2P_k(t)}\int_{-1}^{1}\frac{P_k(s)}{t-s}\,ds.$$

Expanding $1/(t-s)$ at infinity, orthogonality of P_k to $1, s, \ldots, s^{k-1}$ makes the integral $O(t^{-k-1})$. Division by the degree-k polynomial makes the right side $O(t^{-2k-1})$. Multiplying by t and setting $t = 2x+1$ gives the assertion. □

Leading-degree cancellation identifies the next coefficient

For any prefix $a_1, \ldots, a_m$, write

$$G_m(x) = \operatorname*{K}_{j=1}^{m} \frac{a_j}{x} = \frac{p_m(x)}{q_m(x)}.$$

The continuant recurrence is

$$p_m = xp_{m-1} + a_m p_{m-2}, \qquad q_m = xq_{m-1} + a_m q_{m-2},$$

with $(p_{-1}, p_0) = (1, 0)$ and $(q_{-1}, q_0) = (0, 1)$. In particular q_m is monic of degree m, and

$$G_m - G_{m-1} = \frac{(-1)^{m-1}a_1\cdots a_m}{q_m q_{m-1}}. \tag{4}$$

Parity of the continuants shows that G_m is odd. Hence $G_m(x) - G_m(x+1)$ is invariant under $x \mapsto -x-1$; its numerator and denominator in the source's representation are polynomials in $z = x(x+1)$.

Fix $k \geq m+1$, and suppose the first $m-1$ chosen coefficients are $b_1, \ldots, b_{m-1}$. Leave the final coefficient a free and write $G_m(x;a)$. Formula (4) gives

$$G_m(x;a) - B_m(x) = (-1)^{m-1} \left(\prod_{j=1}^{m-1} b_j \right) (a - b_m) x^{-2m+1} + O(x^{-2m-1}). \tag{5}$$

For any $q \geq 1$, a leading term cx^{-q} has difference

$$cx^{-q} - c(x+1)^{-q} = qcx^{-q-1} + O(x^{-q-2}).$$

Moreover, Lemma 1, (2) and (3) show that

$$E_k(x) - \big(B_m(x) - B_m(x+1)\big) = O(x^{-2m-2}), \tag{6}$$

since $k \geq m+1$. Combining (5) and (6) therefore yields

$$E_k(x) - \big(G_m(x;a) - G_m(x+1;a)\big) = (2m-1)(-1)^m \left(\prod_{j=1}^{m-1} b_j \right) (a - b_m) x^{-2m} + O(x^{-2m-1}). \tag{7}$$

The coefficient of $a - b_m$ is nonzero.

Now relate this directly to the polynomial algorithm. With $q_m^+(x) = q_m(x+1)$ and similarly for p_m^+, the cleared numerator is exactly

$$\Delta_1(k,m) = f_k^* q_m q_m^+ - f_k(p_m q_m^+ - p_m^+ q_m).$$

Its denominator is $f_k(z) q_m(x) q_m(x+1)$. As a polynomial in z, this denominator has degree $d_k + m$ and a fixed nonzero leading coefficient. The order estimate (7) thus says that $\Delta_1(k,m)$ has degree at most d_k, and its coefficient of z^{d_k} is a fixed nonzero multiple of $a - b_m$. The source's rule is precisely to set this highest possible coefficient equal to zero. Its unique choice is therefore $a = b_m$. For this choice, the stronger estimate (6) gives $\deg \Delta_1(k,m) \leq d_k - 1$, as required.

Starting with the empty prefix proves the assertion inductively. There is no premature termination before the required last step: after a step $j < m$, (4) and (3) give

$$\mathcal{R}(x) - B_j(x) = (-1)^j \left(\prod_{i=1}^{j+1} b_i \right) x^{-2j-1} + O(x^{-2j-3}).$$

Because $k \geq m+1 \geq j+2$, the Legendre-input error is of strictly higher order than the difference of this expression. Thus the residual has a nonzero term of order x^{-2j-2}; its cleared numerator has degree exactly $d_k - 1$. Every next cancellation above is well defined. This proves Theorem 1. □

Consequences and exact verification

The full stabilized continued fraction is therefore the classical Stirling one. Its standard Stieltjes bounds, already recalled in [1, Section 6], have even convergents below $r(x)$ and odd convergents above $r(x)$, for $x > 0$. This also gives the intended companion bound in MathDB 354156; no separate campaign credit is taken for that immediate consequence. The source's displayed parity inequalities in Conjecture 6.1 are reversed relative to its prose and preceding results; we use the unambiguous intended parity.

The verifier `verify_coefficients.py` implements two independent exact constructions. One builds Legendre polynomials, continuants and cleared residuals, then cancels their leading coefficients. The other starts with the Bernoulli series and repeatedly extracts the continued fraction using formal-series reciprocation. All 45 coefficients in the prefixes $2 \leq k \leq 10$, $1 \leq m < k$, agree. It separately checks the nonstabilized value $a_2^{(2)} = 1/18$. The all-order proof above, not these finite checks, establishes the conjecture.

Tasks 2–5. Further directions after the full solve

These are proposed tasks. Check their current literature and public claims before attacking them.

Task 2. For fixed k, determine whether all coefficients $a_m^{(k)}$ beyond the stabilization region exist and remain positive. Identify the asymptotic behavior of this finite-k tail.

Task 3. Settle the source's separate sign conjecture for all coefficients of the cleared residual $\Delta_1(k, m)$ after its leading-degree cancellation.

Task 4. Derive explicit convergence-rate bounds for the stabilized convergents $B_m(x)$, uniform on $x \geq x_0 > 0$, with usable dependence on both m and x_0.

Task 5. Characterize positive moment sequences whose formal discrete antiderivatives also admit positive Stieltjes continued fractions, extending the structural mechanism behind the Stirling example.

References

[1] G. Bhatnagar and K. Rajkumar, *Telescoping continued fractions for the error term in Stirling's formula*, J. Approx. Theory 293 (2023), 105943, https://doi.org/10.1016/j.jat.2023.105943. Current author-linked manuscript: https://arxiv.org/abs/2204.00962v2. Earlier version, with the stabilization results in Sections 6–7: https://arxiv.org/abs/2204.00962v1.